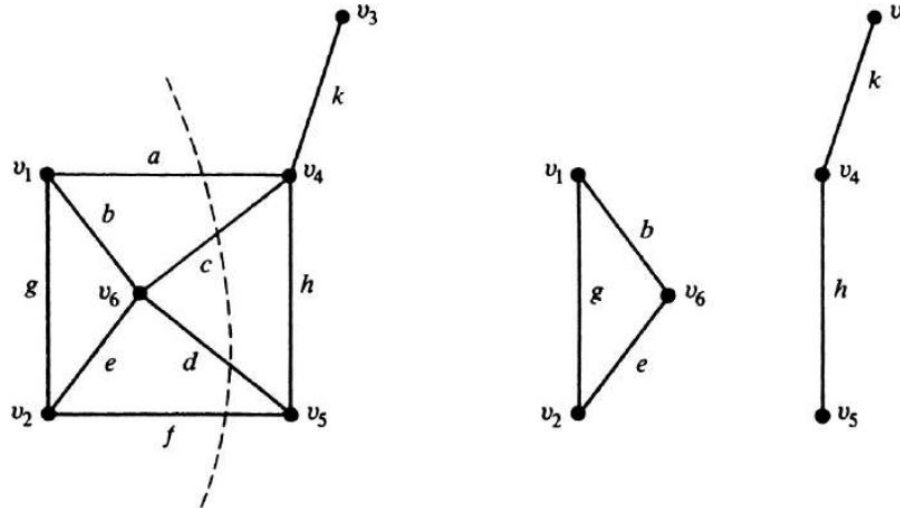


Cutsets and Cutvertices

CUTSETS

In a connected graph G , a **cut-set** is a set of edges whose removal from G leaves G disconnected, provided that removal of no proper subset of these edges disconnects G .

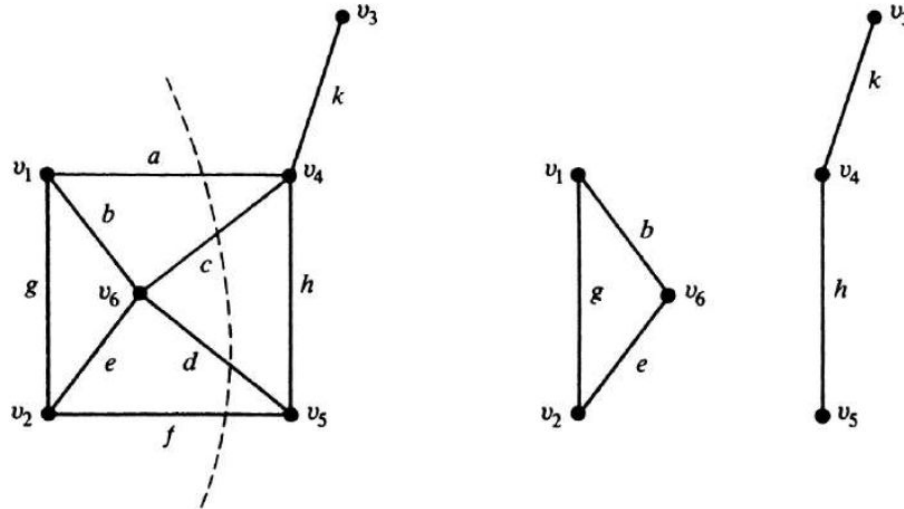
For example, the set of edges $\{a, c, d, f\}$ is a cut-set.



CUTSETS

There are many other cutsets, such as $\{a, b, g\}$, $\{a, b, e, f\}$, and $\{d, h, f\}$. Edge $\{k\}$ alone is also a cut-set.

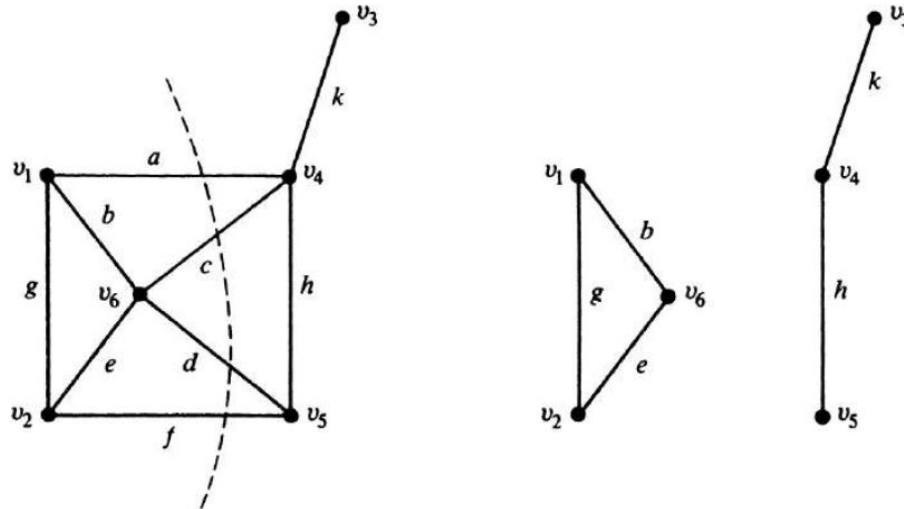
The set of edges $\{a, c, h, d\}$, on the other hand, is not a cut-set, because one of its proper subsets, $\{a, c, h\}$, is a cut-set.



CUTSETS

A cut-set always “cuts” a graph into two.

Therefore, another way of defining a cut-set can be that it is a minimal set of edges in a connected graph whose removal **reduces the rank of the graph by one**.



CUTSETS

Since removal of any edge from a tree breaks the tree into two parts, **every edge of a tree is a cut-set.**

One use of finding cut-sets is immediately obvious, which is **finding weak spots in a network.** If we can find all cutsets of the graph, and the one with the smallest number of edges is the most vulnerable.

PROPERTIES OF A CUT-SET

THEOREM 4-1: *“Every cut-set in a connected graph G must contain at least one branch of every spanning tree of G .”*

THEOREM 4-2: *“In a connected graph G , any minimal set of edges containing at least one branch of every spanning tree of G is a cut-set.”*

The forward direction (Theorem 4.1) proof is quite trivial.

Let's ask ourselves: Is it possible for an arbitrary cut-set S to not have any edge in common with a spanning tree T ? The answer is no.

Otherwise, removal of the cut-set S from G would not disconnect the graph.

PROPERTIES OF A CUT-SET

For the reverse direction (Theorem 4.2), we suppose Q is a minimal set of edges containing at least one branch of every spanning tree of G .

Consider $G - Q$, the subgraph that remains after removing the edges in Q from G . Since the subgraph $G - Q$ contains no spanning tree of G , $G - Q$ is disconnected.

Also, since Q is a minimal set of edges with this property, any edge e from Q returned to $G - Q$ will create at least one spanning tree. Thus the subgraph $G - Q + e$ will be a connected graph.

That means, Q must have been a cut-set, since it was a minimal set of edges whose removal disconnected G .

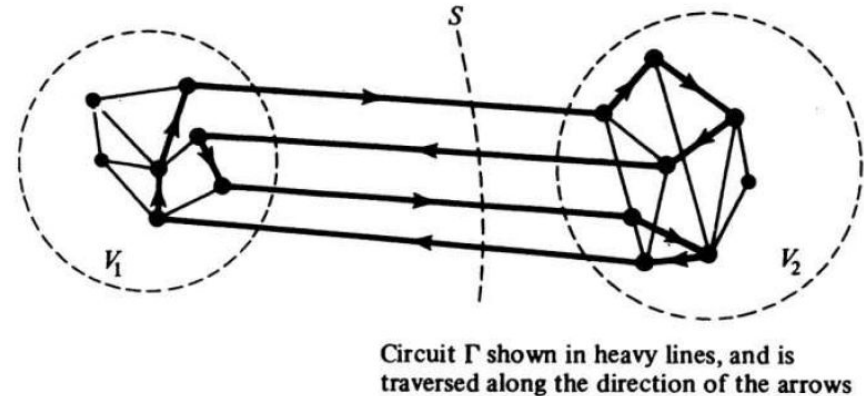
PROPERTIES OF A CUT-SET

THEOREM 4-3: *“Every circuit has an even number of edges in common with any cut-set.”*

Consider a cut-set S in graph G . Since S is a cut-set, removal of S will partition the vertices of G into two (mutually exclusive or disjoint) subsets, say V_1 and V_2 .

Consider a circuit Γ in G .

Case 1: If all the vertices in Γ are entirely within vertex set V_1 (or V_2), the number of edges common to S and Γ is zero; which is an even number.

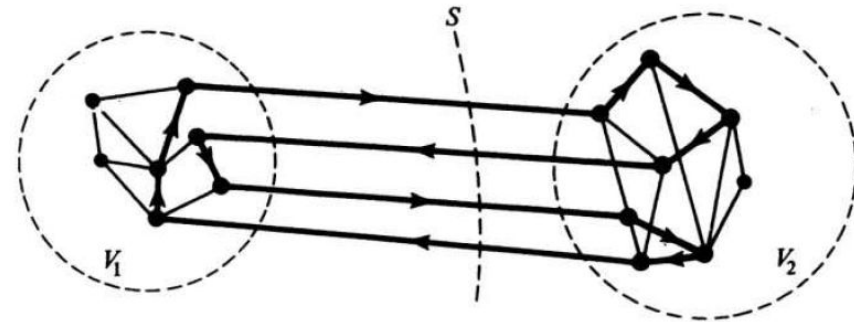


PROPERTIES OF A CUT-SET

THEOREM 4-3: *“Every circuit has an even number of edges in common with any cut-set.”*

Case 2: If, on the other hand, some vertices in Γ are in V_1 and some in V_2 , we traverse back and forth between the sets V_1 and V_2 as we traverse the circuit.

Because of the closed nature of a circuit, the number of edges we traverse between V_1 and V_2 must be even. And since every edge in S has one end in V_1 and the other in V_2 , and no other edge in G has this property, the number of edges common to S and Γ is even.



Circuit Γ shown in heavy lines, and is traversed along the direction of the arrows

FINDING ALL CUTSETS IN A GRAPH

Remember in the previous chapter how we found all circuits in a graph?

We shall follow a similar strategy here.

Just as a spanning tree is essential for defining a set of **fundamental circuits**, so is a spanning tree essential for a set of **fundamental cutsets**.

FINDING ALL CUTSETS IN A GRAPH

Consider a spanning tree T of a connected graph G .

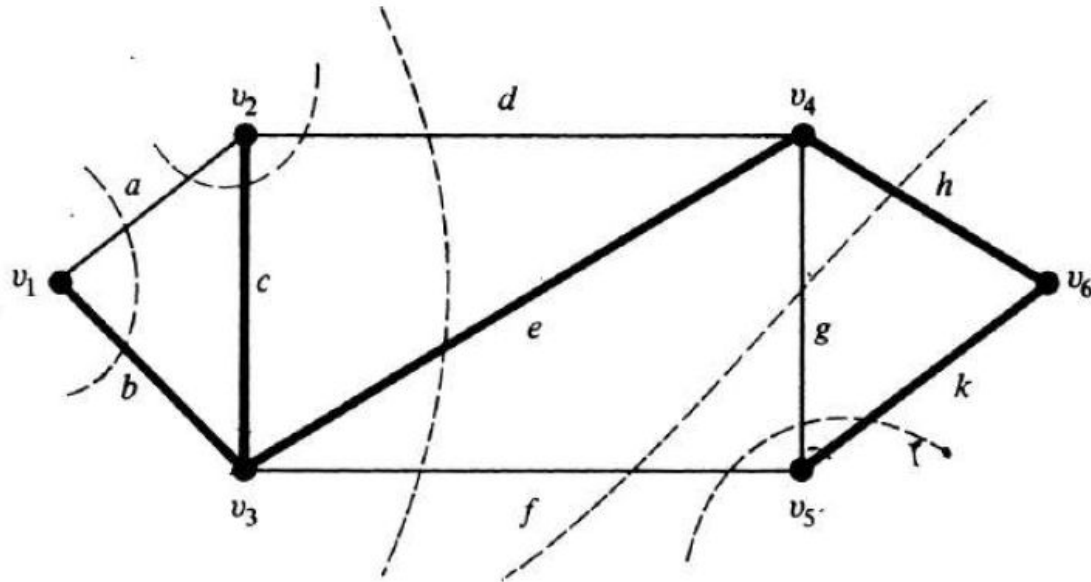
Take any branch b in T . Since $\{b\}$ is a cut-set in T , $\{b\}$ partitions all vertices of T into two disjoint sets—one at each end of b .

Consider the same partition of vertices in the bigger graph G (of which T is a subset), and the cut set S in G that corresponds to this partition. Cutset S will contain only one branch b of T , and the rest (if any) of the edges in S are chords with respect to T .

Such a cut-set S containing exactly one branch of a tree T is called a fundamental cut-set with respect to T .

FINDING ALL CUTSETS IN A GRAPH

For the following graph, a spanning tree T (in heavy lines) and all five of the fundamental cutsets with respect to T are shown.



FINDING ALL CUTSETS IN A GRAPH

Just as every chord of a spanning tree defines a unique fundamental circuit, every branch of a spanning tree defines a unique fundamental cut-set.

Also note that the term fundamental cut-set (like the term fundamental circuit) has meaning **only with respect to a given spanning tree**.

FINDING ALL CUTSETS IN A GRAPH

THEOREM 4-4: *“The ring sum of any two cutsets in a graph is either a third cut-set or an edge-disjoint union of cutsets.”*

Let S_1 and S_2 be two cutsets in a given connected graph G .

Let V_1 and V_2 be the (unique and disjoint) partitioning of the vertex set V of G corresponding to S_1 . Let V_3 and V_4 be the partitioning corresponding to S_2 .

$$V_1 \cup V_2 = V \text{ and } V_1 \cap V_2 = \emptyset,$$

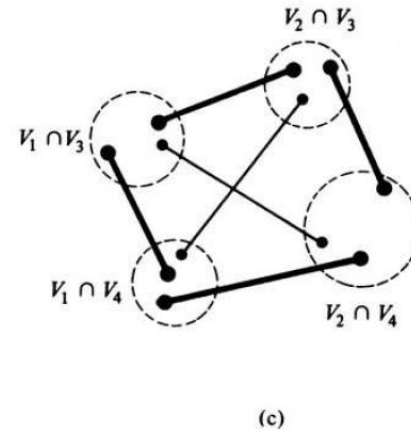
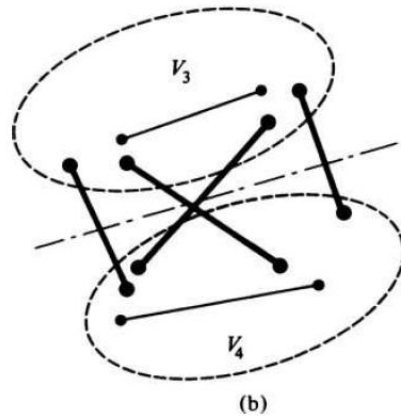
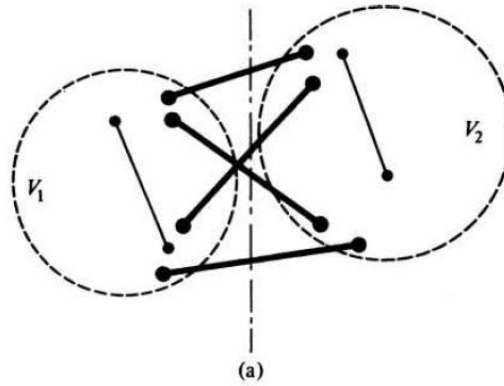
$$V_3 \cup V_4 = V \text{ and } V_3 \cap V_4 = \emptyset.$$

FINDING ALL CUTSETS IN A GRAPH

Now let the subset $(V_1 \cap V_4) \cup (V_2 \cap V_3)$ be called V_5 , and this by definition is the same as the ring sum $V_1 \oplus V_3$.

Similarly, let the subset $(V_1 \cap V_3) \cup (V_2 \cap V_4)$ be called V_6 , which is the same as $V_2 \oplus V_4$.

FINDING ALL CUTSETS IN A GRAPH



FINDING ALL CUTSETS IN A GRAPH

The ring sum of the two cutsets $S_1 \oplus S_2$ can be seen to consist only of edges that join vertices in V_5 to those in V_6 . Also, there are no edges outside $S_1 \oplus S_2$ that join vertices in V_5 to those in V_6 .

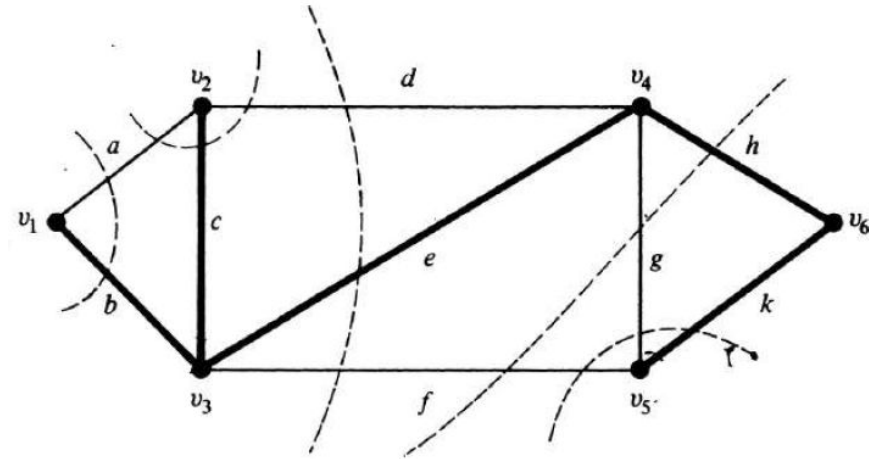
Thus the set of edges $S_1 \oplus S_2$ produces a partitioning of V into V_5 and V_6 such that

$$V_5 \cup V_6 = V \text{ and } V_5 \cap V_6 = \emptyset.$$

Example

Let us consider ring sums of the following three pairs of cutsets.

$$\begin{aligned}\{d, e, f\} \oplus \{f, g, h\} &= \{d, e, g, h\}, && \text{another cut-set,} \\ \{a, b\} \oplus \{b, c, e, f\} &= \{a, c, e, f\}, && \text{another cut-set,} \\ \{d, e, g, h\} \oplus \{f, g, k\} &= \{d, e, f, h, k\} \\ &= \{d, e, f\} \cup \{h, k\}, && \text{an edge-disjoint} \\ &&& \text{union of cut-sets.} \quad \blacksquare\end{aligned}$$



FUNDAMENTAL CIRCUITS AND CUTSETS

From Chapter 3 (Trees), we know that every chord c_i creates a unique fundamental circuit.

Now we are interested in **how many fundamental cutsets contain this chord c_i** . This is answered by the following theorem:

THEOREM 4-5: *“With respect to a given spanning tree T , a chord c_i that determines a fundamental circuit Γ occurs in every fundamental cut-set associated with the branches in Γ and in no other.”*

FUNDAMENTAL CIRCUITS AND CUTSETS

The proof is quite simple. We will make use of **THEOREM 4-3** here.

Consider a spanning tree T where c_i is a chord, and let the fundamental circuit made by c_i be called Γ , consisting of k branches $b_1, b_2, \dots, b_k$ in addition to the chord c_i , that is:

$$\Gamma = \{c_i, b_1, b_2, \dots, b_k\}$$

Every branch of any spanning tree has a different fundamental cut-set associated with it. Let S_1 be the fundamental cut-set associated with b_1 , consisting of q chords in addition to the branch b_1 ; that is:

$$S_1 = \{b_1, c_1, c_2, \dots, c_q\}$$

FUNDAMENTAL CIRCUITS AND CUTSETS

According to **THEOREM 4-3**, there must be an even number of edges common to Γ and S_1 . Edge b_1 is in both Γ and S_1 , and there is only one other edge in Γ (which is c_i) that can possibly also be in S_1 .

Thus the chord c_i is one of the chords $c_1, c_2, \dots, c_q$.

Exactly the same argument holds for fundamental cutsets associated with $b_2, b_3, \dots$, and b_k . Therefore, the chord c_i is contained in every fundamental cut-set associated with branches in Γ .

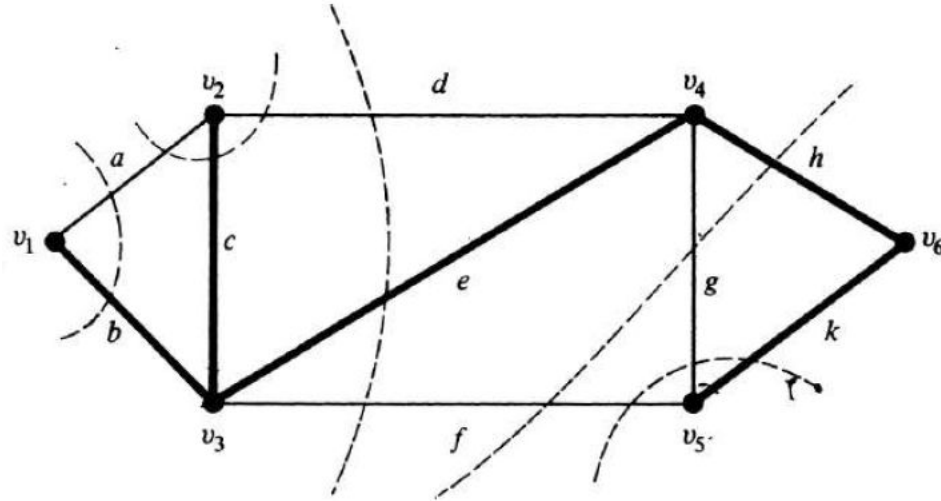
FUNDAMENTAL CIRCUITS AND CUTSETS

Now for the last part of the theorem, is it possible for the chord c_i to be in any other fundamental cut-set S' that is not associated with any of these k branches?

The answer is no. Otherwise as none of the branches in Γ are in S' , there would be only one edge c_i common to S' and Γ , which is an odd number of common edges and thus a contradiction to **THEOREM 4-3**.

FUNDAMENTAL CIRCUITS AND CUTSETS

To see the implications of **THEOREM 4-5**, consider the spanning tree $\{b, c, e, h, k\}$



The fundamental circuit made by chord f is $\{f, e, h, k\}$.

FUNDAMENTAL CIRCUITS AND CUTSETS

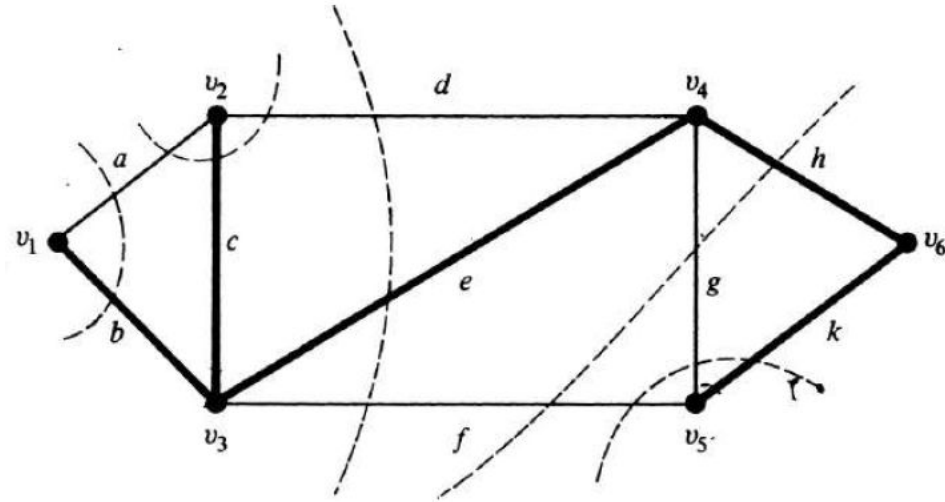
The three fundamental cutsets determined by the three branches e, h, and k are

determined by branch e: {d, e, f},

determined by branch h: {f, g, h},

determined by branch k: {f, g, k}.

Chord f occurs in all 3 fundamental cutsets, and there is no other fundamental cut-set that contains f.



FUNDAMENTAL CIRCUITS AND CUTSETS

The converse of **Theorem 4-5** is also true, which is given as:

THEOREM 4-6: *“With respect to a given spanning tree T , a branch b_i that determines a fundamental cut-set S is contained in every fundamental circuit associated with the chords in S , and in no others.”*

And the proof can be done by the same arguments that led to **THEOREM 4-5**.

CONNECTIVITY AND SEPARABILITY

Edge Connectivity: The number of edges in the smallest cut-set (i.e., cutset with fewest number of edges) is defined as the edge connectivity of G .

It can be alternatively be defined as the minimum number of edges whose removal (i.e., deletion) reduces the rank of the graph by one. The edge connectivity of a tree, for instance, is one.

Vertex Connectivity: The vertex connectivity (or simply connectivity) of a connected graph G is defined as the minimum number of vertices whose removal from G leaves the **remaining graph** disconnected.

Note that from the way we have defined it, vertex connectivity is meaningful only for graphs that have three or more vertices and are not complete.

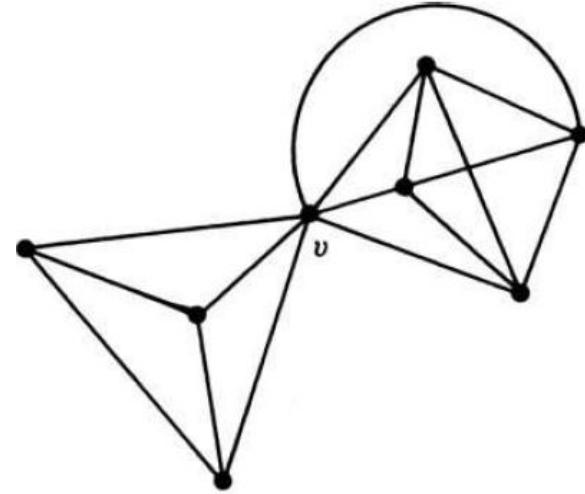
CONNECTIVITY AND SEPARABILITY

Separable Graph: A connected graph is said to be separable if its vertex connectivity is one.

An equivalent definition is that a connected graph G is said to be separable if there exists a subgraph g in G such that g' (the complement of g in G) and g have only one vertex in common.

In a separable graph a vertex whose removal disconnects the graph is called a cut-vertex.

In the graph shown, v is the cut-vertex.



THEOREM 4-7

“A vertex v in a connected graph G is a cut-vertex if and only if there exist two vertices x and y in G such that every path between x and y passes through v .”

The implications of this theorem is quite significant. It implies that v is a crucial vertex in the sense that any communication between x and y (if G represented a communication network) must “pass through” v .

An Application of Connectivity

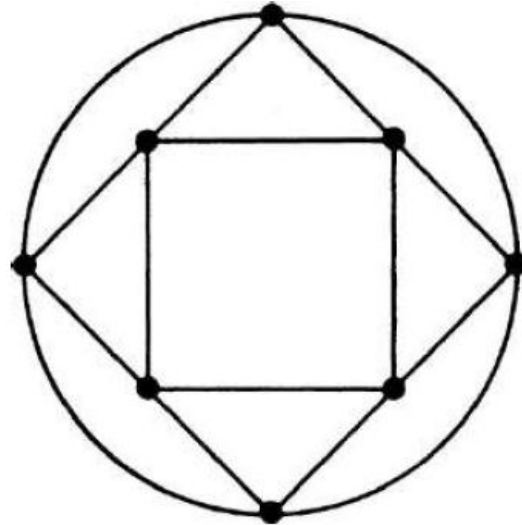
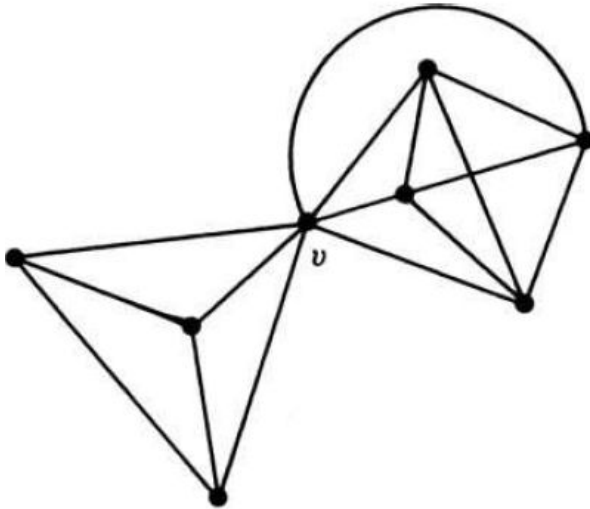
Suppose we are given n stations that are to be connected by means of e lines (telephone lines, bridges, railroads, tunnels, or highways) where $e \geq n - 1$. What is the best way of connecting?

By “best” we mean that the network should be as invulnerable to destruction of individual stations and individual lines as possible.

We can formulate this as a graph theory problem in this way: construct a graph with n vertices and e edges that has the **maximum possible edge connectivity and vertex connectivity**.

An Application of Connectivity

For example, the left graph has $n = 8$, $e = 16$, and has **vertex connectivity of 1** and **edge connectivity of 3**. Another graph with the same number of vertices and edges (8 and 16, respectively) can be drawn as shown in the right graph.



Maximum Vertex and Edge Connectivity

The application we discussed just now leads to the question:

What is the highest vertex and edge connectivity we can achieve for a given n and e ?

Theorems 4-8, 4-9 and 4-10 will constitute the answer to this question.

Maximum Vertex and Edge Connectivity

THEOREM 4-8: “*The edge connectivity of a graph G cannot exceed the degree of the vertex with the smallest degree in G .*”

Let vertex v_i be the vertex with the smallest degree in G .

Let $d(v_i)$ be the degree of v_i . Vertex v_i can be separated from G by removing the $d(v_i)$ edges incident on vertex v_i . Hence the theorem.

Maximum Vertex and Edge Connectivity

THEOREM 4-9: “*The vertex connectivity of any graph G can never exceed the edge connectivity of G .*”

Let α denote the edge connectivity of G .

Therefore, there exists a cutset S in G with α edges. Let S partition the vertices of G into subsets V_1 and V_2 .

By removing at most α vertices from V_1 (or V_2) on which the edges in S are incident, we can effect the removal of S (together with all other edges incident on these vertices) from G . Hence the theorem.

Maximum Vertex and Edge Connectivity

THEOREM 4-10: “*The maximum vertex connectivity one can achieve with a graph G of n vertices and e edges ($e \geq n - 1$) is the integral part of the number $2e/n$; that is, $\lfloor 2e/n \rfloor$.*”

Every edge in G contributes two degrees. The total ($2e$ degrees) is divided among n vertices.

Therefore, there must be at least one vertex in G whose degree is equal to or less than the number $2e/n$.

The vertex connectivity of G cannot exceed this number, in light of **Theorems 4-8** and **4-9**.

Maximum Vertex and Edge Connectivity

The results of Theorems 4-8, 4-9, and 4-10 can be summarized as follows:

$$\textit{vertex connectivity} \leq \textit{edge connectivity} \leq \left\lfloor \frac{2e}{n} \right\rfloor$$

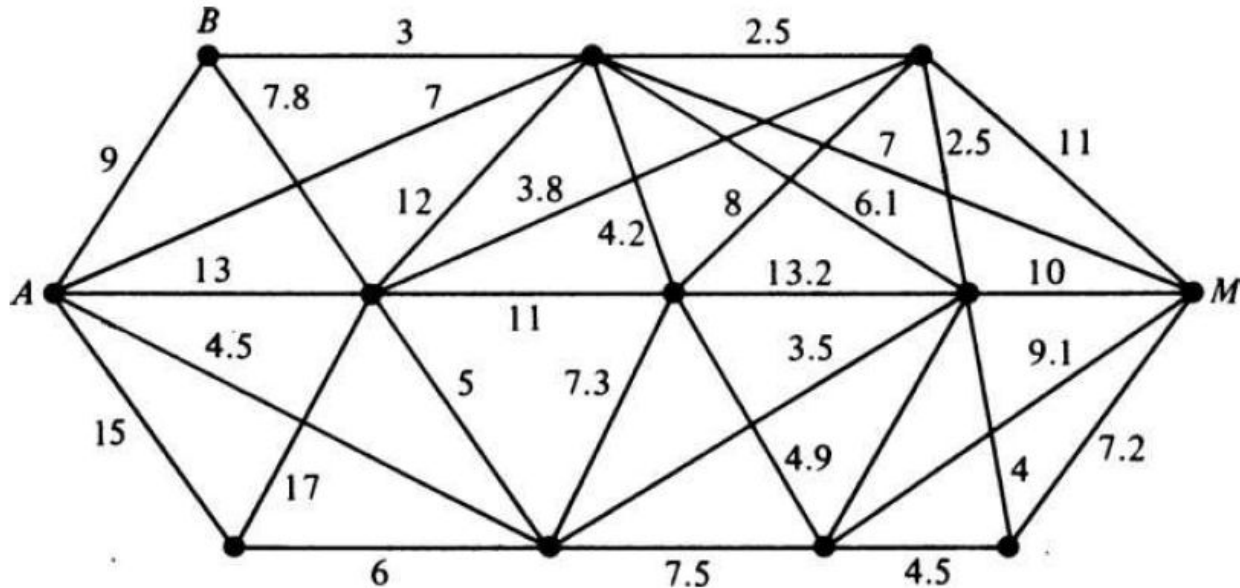
$$\max(\textit{vertex connectivity}) = \left\lfloor \frac{2e}{n} \right\rfloor$$

Thus, for a graph with 8 vertices and 16 edges, we can achieve a vertex connectivity (and therefore edge connectivity) as high as 4 ($=2 \cdot 16/8$).

A graph G is said to be **k-connected** if the vertex connectivity of G is k ; therefore, a 1-connected graph is the same as a separable graph.

NETWORK FLOWS

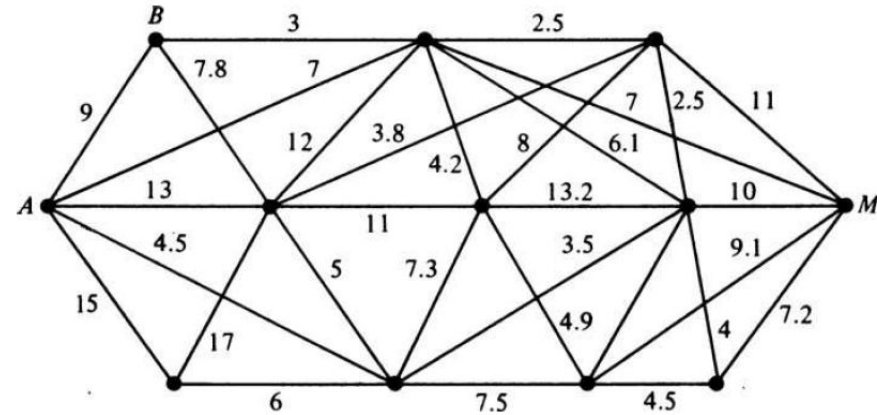
A **flow network** is a **weighted connected graph** where each edge has a weight representing its **capacity** and each edge receives a flow. The amount of flow on an edge cannot exceed the **capacity** of the edge.



NETWORK FLOWS

In such a flow problem the questions to be answered are

1. What is the maximum flow possible through the network between a specified pair of vertices, say, from **B** to **M** in this graph?
2. How do we achieve this flow (i.e., determine the actual flow through each edge when the maximum flow exists)?



Max Flow - Min Cut

THEOREM 4-13: *“The maximum flow possible between two vertices a and b in a network is equal to the minimum of the capacities of all cutsets with respect to a and b .”*

Consider any cut-set S that separates vertices a and b in G .

In the subgraph $G - S$ (the subgraph left after removing S from G) there is no path between a and b . Therefore, every path in G between a and b must contain at least one edge of S .

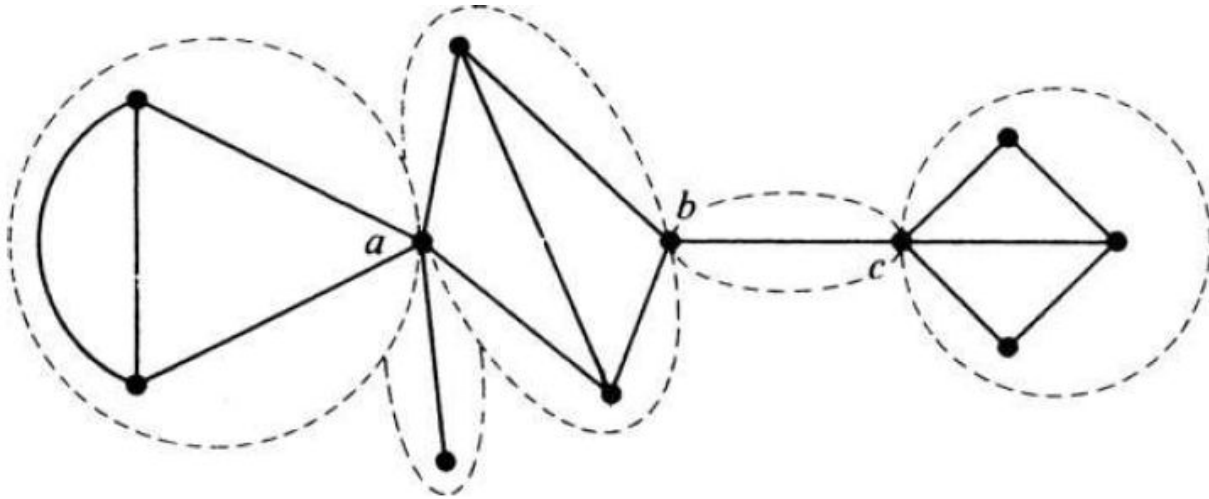
Thus every flow from a to b (or from b to a) must pass through one or more edges of S . Hence the total flow rate between these two vertices cannot exceed the capacity of S .

Since this holds for all cutsets with respect to a and b , the flow rate cannot exceed the minimum of their capacities.

1-ISOMORPHISM

A separable graph (with vertex connectivity of 1) consists of two or more nonseparable subgraphs. Each of the largest nonseparable subgraphs is called a **block**.

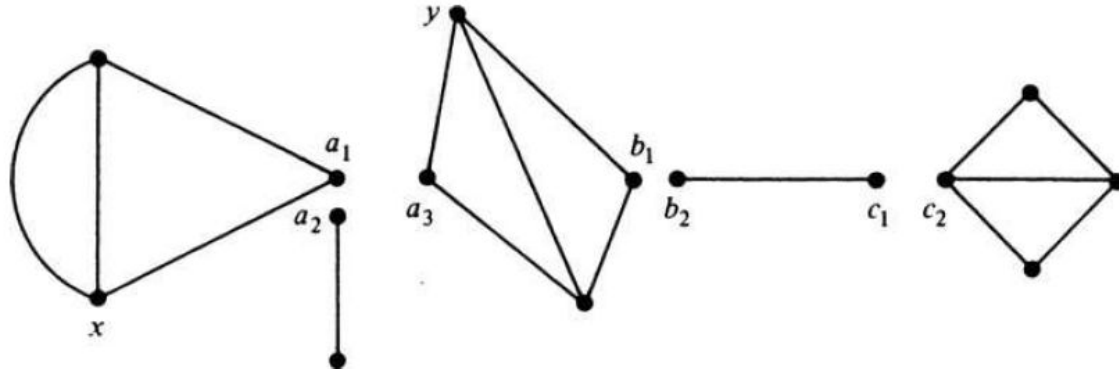
The following graph has 5 blocks (and three cut-vertices a , b and c).



1-ISOMORPHISM

Now, visually compare the disconnected graph shown here with the one shown in the previous slide.

These two graphs are certainly not isomorphic (they do not have the same number of vertices), but they are related by the fact that the **blocks** of the previous graph are isomorphic to the **components** of this graph. Such graphs are said to be **1-isomorphic**.



1-ISOMORPHISM

Two graphs G_1 and G_2 are said to be **1-isomorphic** if they become **isomorphic** to each other under repeated application of the following operation.

Operation 1: “Split” a cut-vertex into two vertices to produce two disjoint subgraphs.

For two nonseparable graphs, **isomorphism** and **1-isomorphism** are the same.

1-ISOMORPHISM

THEOREM 4-14: *“If G_1 and G_2 are two 1-isomorphic graphs, the rank of G_1 equals the rank of G_2 and the nullity of G_1 equals the nullity of G_2 .”*

Under **operation 1**, whenever a cut-vertex in a graph G is “split” into two vertices, the number of components (k) in G increases by one. But the split vertex also becomes two vertices, so the number of vertices (n) also increases by 1.

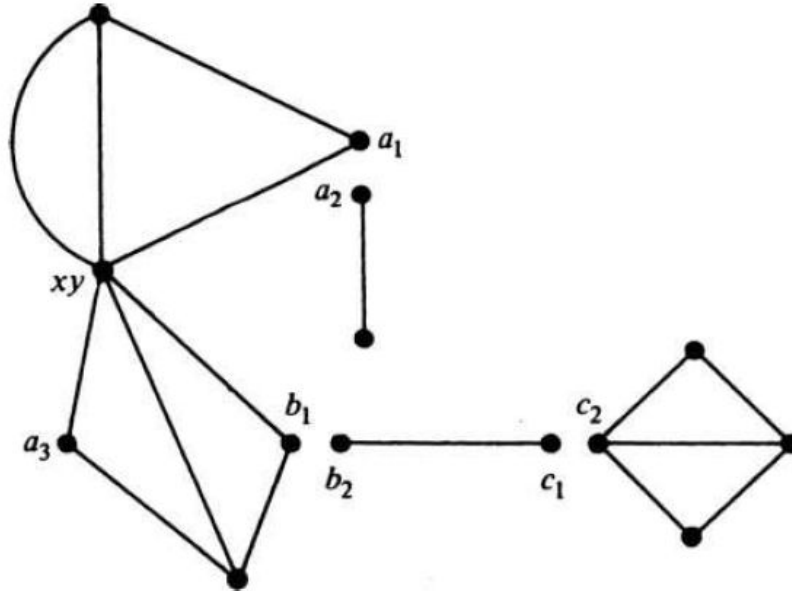
Therefore, the rank ($n-k$) remains invariant under **operation 1**.

Also, since no edges are destroyed or new edges created by operation 1, two 1-isomorphic graphs have the same number of edges. Two graphs with equal rank and with equal numbers of edges must have the same nullity, because

$$\text{nullity} = \text{number of edges} - \text{rank}$$

1-ISOMORPHISM

If we “glue” together different vertices in the previous disconnected graph, such that the blocks are still the same, then that differently “glued” graph will also be 1-isomorphic to the previous two graphs. For example,



2-ISOMORPHISM

Graphs that are **isomorphic** are **also 1-isomorphic**, but **1-isomorphic** graphs may **not be isomorphic**. This generalized isomorphism is very useful in the study of separable graphs.

We can generalize this concept further to broaden its scope for 2-connected graphs (i.e., graphs with vertex connectivity of two).

2-ISOMORPHISM

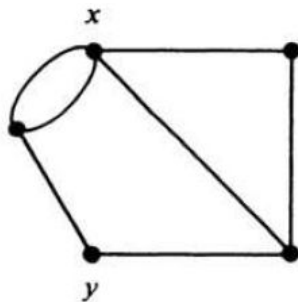
In a 2-connected graph G let vertices x and y be a pair of vertices whose removal from G will leave the remaining graph disconnected.

Suppose that we perform the following **operation 2** on G (after which, of course, G no longer remains the original graph).

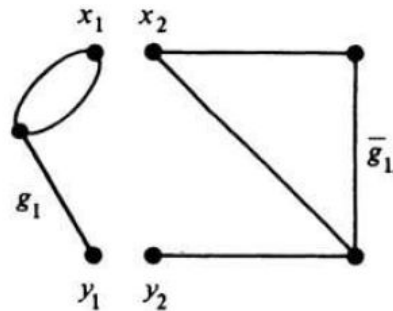
Operation 2: “Split” the vertex x into x_1 and x_2 and the vertex y into y_1 and y_2 such that G is split into g_1 and g_2 . Let vertices x_1 and y_1 go with g_1 and x_2 and y_2 with g_2 . Now rejoin the graphs g_1 and g_2 by merging x_1 with y_2 and x_2 with y_1 .

Two graphs are said to be **2-isomorphic** if they become **isomorphic** after undergoing **operation 1** or **operation 2**, or both operations any number of times.

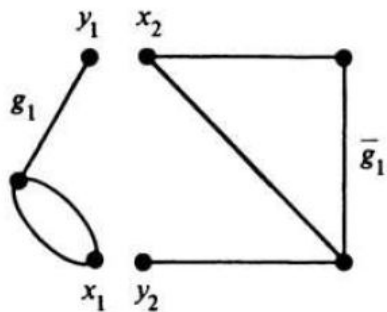
2-ISOMORPHISM



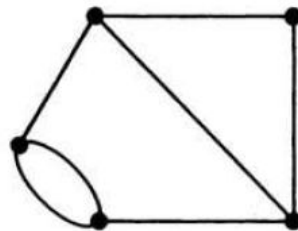
(a)



(b)



(c)



(d)

END OF LECTURE